

Gamification in Education

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1 Introduction

Once, we learned for knowledge.

Then came the points.

Then the badges.

Then the leaderboards.

And suddenly, education was no longer about learning.

It was about XP.

Gamification has become increasingly common in education, with points, badges, levels and rewards being used to motivate students and increase engagement. For Ester, however, gamification is not something she is particularly fond of. Having worked in a school and as a private tutor, Ester has seen how deeply gamification can influence the learning experience. Her father is also a teacher and frequently talks about the use of different methods to engage students. Even in her own professional learning, she has encountered gamification while preparing for the Salesforce Administrator exam, where challenges, badges and progress systems are heavily integrated. Gamification is therefore no longer limited to games. It has become part of education, professional development and everyday learning. This raises an important question: does gamification actually improve learning, or does it simply make education more engaging? Reviews of the field suggest that gamification does have positive cognitive, emotional and behavioural effects overall, but that the quality of the design and the specific game elements used matter as much as the presence of gamification itself [1].

2 The Battle of the Gamers

Ester is sitting at home, the room lit only by the pale glow of her laptop screen. On the desk beside her, a can of soda has gone flat, forgotten somewhere between the third and fourth round. The notification sound plays again, a short bright chime, the same chime that plays every time someone's name moves. She has learned to tell the difference between the chime that means someone climbed

and the one that means someone fell, though the sound itself never changes, only her stomach does. On the screen, a leaderboard refreshes itself in real time. Names rearrange, numbers tick upward in green, small animated stars burst beside whoever sits at the top. Her own name sits near the bottom, grey against all that shine, and no matter how quickly she answers, no matter how many rounds she plays, it barely moves. Through the wall she can hear her brother in the next room, laughing, his keyboard clattering fast and easy, his name climbing without him even seeming to try. Downstairs, her mother calls up to ask how she is doing, and Ester types back that she is doing great, thumbs quick over the keys, the lie smoother than she expected it to be. Her palms are damp against the trackpad. Her jaw aches from smiling at a screen no one else can see. Every few minutes a new round begins, and the timer counts down in the corner, red numbers shrinking second by second, and her hand hovers over the keys, already bracing for the small drop her name will take before the numbers even land.

References

- [1] Rabah, Cassidy, and Beauchemin, “Gamification in education: Real benefits or edutainment?,” in *European Conference on eLearning (ECEL)*, 2018.
- [2] Gomez, Moron, Ferrandez, Saiz, and Garrido, “Gamification in the education: The benefits of learning through play,” in *INTED2017 Proceedings*, IATED, 2017.
- [3] Chapman and Rich, “Does educational gamification improve students’ motivation? if so, which game elements work best?,” *Journal of Education for Business*, 2018.
- [4] Nistor and Iacob, “The advantages of gamification and game based learning and their benefits in the development of education,” in *International Scientific Conference eLearning and Software for Education (eLSE)*, 2018.
- [5] Romano Daba, Rosmansyah, and Dabarsyah, “Problem based learning using gamification: A systematic literature review,” in *IEEE International Conference on Information and Communication Technology and Systems (ICIMCIS)*, 2019.
- [6] Harviainen, “Critical challenges to gamifying education: A review of central concepts,” in *Game On! Conference*, (Moscow), 2014.
- [7] Biryukov, Brikoshina, Mikhalevich, Sycheva, and Khalimon, “Gamification in education: threats or new opportunities,” in *SHS Web of Conferences*, 2021.
- [8] Mirzaie Feiz Abadi, Khalili Samani, Akhlaghi, Najibi, and Bolourian, “Pros and cons of tomorrow’s learning: A review of literature of gamification in education context,” *Medical Education Bulletin*, 2022.

- [9] Toda, Valle, and Isotani, “The dark side of gamification: An overview of negative effects of gamification in education,” in *Higher Education for All (HEFA) Workshop*, Communications in Computer and Information Science, Springer, 2018.
- [10] Thorsteinsson and Niculescu, “Pedagogical insights into the use of minecraft within educational settings,” *Studies in Informatics and Control*, 2016.
- [11] Slattey, Butler, O’Leary, and Marshall, “Primary school students’ experiences using Minecraft education during a national project based initiative: An irish study,” *TechTrends*, 2023.
- [12] Baek, Min, and Yun, “Mining educational implications of Minecraft,” *Computers in the Schools*, 2020.
- [13] Kuutti, “Designing gamification,” Master’s thesis, Oulu Business School, 2016.
- [14] Steinbeiss, “Minecraft as a learning and teaching tool: Designing integrated game experiences for formal and informal learning activities,” Master’s thesis, University of Oulu, 2017.
- [15] Caponetto, Earp, and Ott, “Gamification and education: A literature review,” in *8th European Conference on Games Based Learning (ECGBL)*, 2014.
- [16] Christopoulos and Mystakidis, “Gamification in education,” *Encyclopedia*, 2023.
- [17] Dicheva, Dichev, Agre, and Angelova, “Gamification in education: A systematic mapping study,” *Educational Technology & Society*, 2015.
- [18] Huang and Soman, “A practitioner’s guide to gamification of education,” tech. rep., Behavioural Economics in Action, Rotman School of Management, University of Toronto, 2013.
- [19] Ofosu Ampong, “The shift to gamification in education: A review on dominant issues,” *Journal of Educational Technology Systems*, 2020.
- [20] Kim and Werbach, “More than just a game: Ethical issues in gamification,” *Ethics and Information Technology*, 2016.
- [21] Khoshnoodifar, Ashouri, and Taheri, “Effectiveness of gamification in enhancing learning and attitudes: A study of statistics education for health school students,” *Journal of Advances in Medical Education and Professionalism*, 2023.